TASK 3: AUTOMATION SYSTEM DESIGN

SMART HOME LIGHTING SYSTEM

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DATE	02/10/2026
Topic	Smart Home Lighting System
Design Type	Automatic lighting control using PIR + LDR sensors

1. Introduction

A Smart Home Lighting System is an automated system that controls indoor lighting according to room occupancy and ambient light conditions. The proposed design uses a Passive Infrared (PIR) sensor to detect human movement, an LDR (Light Dependent Resistor) to determine whether the room is sufficiently dark, and a microcontroller to make the control decision. A relay driver then switches the lamp safely.

The main objective is to provide lighting when it is actually required while reducing unnecessary energy consumption. The system can operate automatically and can also include a manual override switch for user control.

2. Problem Statement

Conventional lights are often switched manually. A light may remain ON when a person leaves the room, or it may be switched ON during daylight when artificial lighting is not required. This causes avoidable electricity consumption and reduces convenience**.**

The proposed automation solves this by combining two conditions: darkness and human presence. The lamp is switched ON when the room is dark, and motion is detected. If the room is bright or no motion is detected for the preset delay, the lamp remains OFF or is switched OFF.

3. Components Required

Component	Purpose
PIR sensor	Detects human movement/presence
LDR sensor	Measures ambient light level
Arduino / ESP32	Processes sensor signals and controls the output
Relay module	Switches the lamp/load from the controller output.
LED lamp / bulb	Provides illumination.
5 V regulated supply	Provides low-voltage power to the control circuit
Manual switch	Optional ON/AUTO/OFF override

4. Working Principle

The system starts by powering the sensors and microcontroller. The LDR is continuously monitored to estimate the ambient light level. The controller compares this value with a selected threshold. If the room is bright, the lamp stays OFF. If the room is dark, the controller checks the PIR sensor.

When motion is detected in the dark room, the microcontroller activates the relay driver and the lamp turns ON. A timer can be used to keep the lamp ON for a selected period. If no additional motion is detected during the delay period, the controller deactivates the relay and the lamp turns OFF. The sensing cycle then repeats.

5. Automation Flow

The design follows this sequence,

• **Start**: Initialize the microcontroller, PIR sensor, LDR,and relay output.

• **Check ambient light**: Read the LDR value and determine whether the room is dark.

• **Check occupancy**: If the room is dark, read the PIR sensor for motion.

• **Switch ON**: When motion is detected, activate the relay and turn the lamp ON.

• **Delay and recheck**: Keep monitoring for movement. Reset the timer whenever a new motion is detected.

• **Switch OFF**: If no motion is detected for the present time, deactivate the relay and turn on the lamp OFF.

• **Repeat**: Continue the sensing and control cycle automatically.

7. Conclusion

The Smart Home Lighting System demonstrates a practical automation application using sensors, microcontrollers, and an actuator. By combining ambient-light detection with human-motion detection, the system can make a simple lighting decision automatically. The separate design file shows the signal flow from the sensors through the controller and relays to the lamp, together with the control logic used by the system.

THANK YOU